

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III

Topics in Analysis (3TN)

Revision on Metric spaces

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0.1 Preliminaries on Metric Spaces

Our intention in this lecture is not about revising the module on Metric Spaces. We would like to quickly revisit some of those concepts which play important roles in Mathematical Analysis and which will be needed throughout these lectures. For instance, the notion of compactness is very important when establishing existence results in Mathematical Analysis.

Definition 0.1 Let X be a nonempty set. We say that $d : X \times X \rightarrow \mathbb{R}$ is a distance if

- (i) $\underbrace{d(x, y) \geq 0 \quad \forall x, y \in X}_{\text{positivity}} \quad \text{and} \quad \underbrace{d(x, y) = 0 \Leftrightarrow x = y}_{\text{vanishing property}};$
- (ii) $\underbrace{d(x, y) = d(y, x) \quad \forall x, y \in X}_{\text{Symmetry}};$
- (iii) $\underbrace{d(x, y) \leq d(x, z) + d(z, y) \quad \forall x, y, z \in X}_{\text{Triangle inequality}}.$

We then say that (X, d) is a metric space.

Example 0.2

- (i) $(\mathbb{R}, d)$ where $d(x, y) = |x - y|$.
- (ii) $(\mathbb{C}, d)$ where $d(w, z) = |w - z|$ the modulus of the complex number $w - z$.
- (iii) $(\mathbb{R}^n, d_2), (\mathbb{R}^n, d_1), (\mathbb{R}^n, d_\infty)$ where

$$d_2(x, y) : = \left(\sum_{k=1}^n (x_k - y_k)^2 \right)^{1/2} \quad (\text{the Euclidean distance}),$$

$$d_1(x, y) : = \sum_{k=1}^n |x_k - y_k|$$

$$d_\infty(x, y) : = \max_{1 \leq k \leq n} |x_k - y_k|.$$

- (iv) (X, d_0) where X is any non empty set and $d_0(x, y) = \begin{cases} 1 & \text{if } x \neq y, \\ 0 & \text{if } x = y. \end{cases}$
 d_0 is called the discrete distance.

- (v) $C^0([a, b], d_\infty)$ where $C^0([a, b])$ is the vector space of the function $f : [a, b] \rightarrow \mathbb{R}$ continuous and d_∞ is defined by

$$d_\infty(f, g) = \max_{a \leq x \leq b} |f(x) - g(x)|.$$

Let (X, d) be a metric space. We are going to assume that you remember the notations and definitions:

1. $B[x, r]$ and $B(x, r)$ of the closed and open ball in X centered at x with radius $r > 0$ respectively.
2. $\text{Int}(E) = E^\circ$, E' and $\bar{E}$ are the interior, the set of accumulation points, the closure of the subset E of X respectively.
3. The diameter of the set E is defined as follows

$$\text{diam}(E) := \begin{cases} \sup_{(x,y) \in E \times E} d(x, y) & \text{if } E \neq \emptyset, \\ 0 & \text{if } E = \emptyset. \end{cases}$$

Notice that $E \subset X$ is bounded if and only if $\text{diam}(E) < +\infty$. **Show that this is equivalent to the fact there exists $x \in X$ and $r > 0$ such that $E \subset B[x, r]$.**

4. We say that the set $E \subset X$ is dense in X if $X = \bar{E}$. That is, every $x \in X$ is a limit of some sequence $(x_n) \subset E$. We recall that the metric space (X, d_X) is separable if X contains a countable dense subset. For instance $\mathbb{R}$ equipped with the usual Euclidean distance is separable. (Why?).
5. We recall that the metric space (X, d_X) is complete if every Cauchy sequence $(x_n) \subset X$ converges in X .
We say that a subset E of X is complete if $(E, d|_{E \times E})$ is a complete metric space.
6. Let (X, d_X) and (Y, d_Y) be two metric spaces and $E \subset X$. Let $f : E \rightarrow Y$ be a mapping.
 - (a) We say that f is bounded if $f(E)$ is a bounded subset of (Y, d_Y) .
 - (b) We say that ℓ is the limit of f at $x \in \bar{E}$ if for every $\varepsilon > 0$, there exists $\delta_{\varepsilon, x} > 0$ such that

$$y \in B_X(x, \delta_{\varepsilon, x}) \cap E \Rightarrow f(y) \in B_Y(\ell, \varepsilon).$$

Explain why $B_X(x, \delta_{\varepsilon, x}) \cap E \neq \emptyset$!

We say that f is continuous at $x \in E$ if $\ell = f(x)$. We say that f is continuous on $A \subset E$ if f is continuous at every $x \in A$.

- (c) We say that the mapping $f : X \rightarrow Y$ is a homeomorphism if f is bijective and f and f^{-1} are continuous. In that case, we say that the spaces (X, d_X) and (Y, d_Y) are homeomorphic.
- (d) We say that $f : E \rightarrow Y$ is *uniformly continuous* if

$$\forall \varepsilon > 0 \quad \exists \delta_\varepsilon > 0 \text{ such that } \forall x, y \in E \quad d_X(x, y) < \delta_\varepsilon \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

- (e) We say that $f : E \rightarrow Y$ is an *hölderian function with exponent* $\alpha \in (0, 1)$ or α -*hölderian* if there exists $C > 0$ such that

$$d_Y(f(x), f(y)) \leq C(d_X(x, y))^\alpha \quad \forall x, y \in E.$$

- (f) We say that f is a *Lipschitz function* if there exists $K > 0$ such that

$$d_Y(f(x), f(y)) \leq K d_X(x, y) \quad \forall x, y \in E. \quad (*)$$

The smallest constant K such that $(*)$ holds is defined by

$$\text{lip}(f, E, Y) := \sup \left\{ \frac{d_Y(f(x), f(y))}{d_X(x, y)} : x, y \in E \text{ with } x \neq y \right\}.$$

and is called the *Lipschitz constant* of f .

- (g) We say that $f : X \rightarrow X$ is a *contraction* if f is Lipschitz with Lipschitz constant in $[0, 1)$.
- (h) We say that $f : E \rightarrow Y$ is an *isometry* if $d_Y(f(x), f(y)) = d_X(x, y)$ for all $x, y \in E$.
- (i) We recall that two distances d_1 and d_2 on X are topologically equivalent if for every $x \in X$ and for every $\varepsilon > 0$, there exists $\delta_{\varepsilon, x} > 0$ such that

$$B_{d_1}(x, \delta_{\varepsilon, x}) \subset B_{d_2}(x, \varepsilon) \quad \text{and} \quad B_{d_2}(x, \delta_{\varepsilon, x}) \subset B_{d_1}(x, \varepsilon).$$

That is, the identity mapping $\text{Id} : (X, d_1) \rightarrow (X, d_2)$ is a homeomorphism.

- (j) We recall that two distances d_1 and d_2 on X are equivalent if there exist $C_1 > 0$ and $C_2 > 0$ such that

$$C_1 d_1(x, y) \leq d_2(x, y) \leq C_2 d_1(x, y) \quad \forall x, y \in X.$$

That is, The identity mapping $\text{Id} : (X, d_1) \rightarrow (X, d_2)$ and its inverse $\text{Id} : (X, d_2) \rightarrow (X, d_1)$ are Lipschitz.

0.2 Contraction mapping theorem

0.2.1 The theorem

The concept of contraction in metric spaces as defined in (g) plays an important role in Mathematical Analysis as it leads to one of the most famous fixed point theorem due to *Stefan Banach* (Polish mathematician, 30 March 1892 - 31 August 1945).

Given $f : (X, d) \rightarrow (X, d)$. We recall that $x \in X$ is a fixed point for the mapping f if $f(x) = x$.

Theorem 0.3 (Contraction mapping theorem)

Let (X, d) be a metric space and let $f : X \rightarrow X$ be a mapping. If the metric space (X, d) is complete and the mapping f is a contraction, then f has a unique fixed point, i.e., there exists a unique point $x \in X$ such that $f(x) = x$. Moreover, for any $x_0 \in X$, the recurrent sequence (x_n) defined by

$$x_{n+1} = f(x_n), \quad n \in \mathbb{N}$$

converges to the fixed point x .



Figure 1: **Stefan Banach: Polish mathematician, 30 March 1892 - 31 August 1945.**

Proof:

Uniqueness: Let $y \in X$ be also such that $f(y) = y$, then we get

$$\begin{aligned} d(x, y) &= d(f(x), f(y)) \leq Kd(x, y) \text{ with } K \in (0, 1) \Rightarrow 0 \leq (1 - K)d(x, y) \leq 0 \\ \Rightarrow d(x, y) &= 0 \Rightarrow x = y. \end{aligned}$$

Existence: We just need to show that the sequence (x_n) converges and that its limit is a fixed point. Notice that

$$d(x_{j+1}, x_j) = d(f(x_j), f(x_{j-1})) \leq Kd(x_j, x_{j-1}) \leq \dots \leq K^j d(x_1, x_0) \quad \forall j \in \mathbb{N}^*.$$

Hence, for every $n, p \in \mathbb{N}$, we get

$$\begin{aligned} d(x_{n+p}, x_n) &\leq \sum_{j=1}^p d(f(x_{n+j}), f(x_{n+j-1})) \leq \sum_{j=1}^p K^{n+j-1} d(x_1, x_0) = K^n d(x_1, x_0) \sum_{j=1}^p K^{j-1} \\ &\leq K^n d(x_1, x_0) \sum_{j=1}^{\infty} K^{j-1} = \frac{K^n}{1-K} d(x_1, x_0). \end{aligned}$$

Therefore,

$$\lim_{n \rightarrow \infty} d(x_{n+p}, x_n) = 0 \quad \text{which implies that } (x_n) \text{ is a Cauchy sequence in } (X, d).$$

Since (X, d) is complete by assumption, it follows that the sequence (x_n) converges in (X, d) to some point $x \in X$. That is, $\lim_{n \rightarrow \infty} d(x_n, x) = 0$. Now, since (x_{n+1}) is a subsequence of (x_n) , it follows that $\lim_{n \rightarrow \infty} d(x_{n+1}, x) = 0$. Now,

$$d(x_{n+1}, f(x)) = d(f(x_n), f(x)) \leq Kd(x_n, x) \rightarrow 0 \Rightarrow \lim_{n \rightarrow \infty} d(x_{n+1}, f(x)) = 0.$$

Hence, we obtain

$$\begin{cases} \lim_{n \rightarrow \infty} d(x_{n+1}, x) = 0, \\ \lim_{n \rightarrow \infty} d(x_{n+1}, f(x)) = 0 \end{cases} \Rightarrow f(x) = x \text{ (by the uniqueness of the limit).}$$

Therefore, the limit x is a fixed point for the mapping f and this achieves the proof of Theorem 0.3. ■

0.2.2 An application of the contraction mapping theorem

We explore in this section, a well-known application of the contraction mapping theorem in the literature. The goal here is not to find general results but rather illustrate some techniques for the application of the abstract result above. The application of the contraction mapping theorem here is the well-known Cauchy-Lipschitz-Picard theorem on the local existence result for the Cauchy problem or the initial value problem for ordinary differential equations.

Theorem 0.4 (Cauchy-Lipschitz-Picard)

Let $a < b$ and let $x_0 \in (a, b)$. Let $f : [a, b] \times \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$\exists K > 0 \text{ such that } |f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2| \quad \forall y_1, y_2 \in \mathbb{R}, \forall x \in [a, b]. \quad (1)$$

Then, there exist $\delta > 0$ and a unique function $y : I[x_0, \delta] \rightarrow \mathbb{R}$ of class $\mathcal{C}^1$ such that

$$\begin{cases} y'(x) = f(x, y(x)), \\ y(x_0) = y_0. \end{cases} \quad (2)$$

Proof: Let us first notice that y solves (2) if y solves the equation

$$y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt \quad (3)$$

In fact, y solves (2), then integrating the ODE, using the initial condition, we get (3). On the other hand, if y solves (3), then we easily obtain the initial condition $y(x_0) = y_0$ and by the fundamental theorem of calculus, get that

$$y'(x) = \frac{d}{dx} \left(y_0 + \int_{x_0}^x f(t, y(t)) dt \right) = f(x, y(x)).$$

Therefore, we are going to show that the equation (3) has a unique solution. We are going to find the solution of (3) through an application of the contraction mapping theorem to the mapping defined by the left-hand side of (3).

Let $\beta > 0$ and let

$$M := \sup_{x \in [a, b] \times I[y_0, \beta]} |f(t, s)| \quad \text{and} \quad \delta := \min \left(b - a, \frac{\beta}{M}, \frac{1}{K} \right). \quad (4)$$

Set

$$X := \mathcal{C}(I[x_0, \delta], I[y_0, \beta]).$$

X equipped with the distance

$$d_\infty(f, g) := \sup_{x \in I[(x_0, \delta)]} |f(x) - g(x)|$$

is a **complete metric space**. We consider the mapping $\Phi : X \rightarrow X$ defined by

$$\Phi(y)(x) := y_0 + \int_{x_0}^x f(t, y(t)) dy \quad \forall y \in X, \forall x \in I[x_0, \delta]. \quad (5)$$

- (i) **Φ is well-defined**, i.e., $\Phi(y) \in X$ for every $y \in X$. In fact,

$$|\Phi(y)(x) - y_0| = \left| \int_{x_0}^x f(t, y(t)) dt \right| \leq M\delta \leq \beta \quad \forall y \in X, \forall x \in I[x_0, \delta].$$

- (ii) **The mapping Φ is a contraction**. In fact, let $y_1, y_2 \in X$ and let $x \in I[(x_0, \delta]$, we have

$$\begin{aligned} |\Phi(y_1)(x) - \Phi(y_2)(x)| &= \left| \int_{x_0}^x [f(t, y_1(t)) - f(t, y_2(t))] dt \right| \leq K\delta d_\infty(y_1, y_2) \\ \Rightarrow d_\infty(\Phi(y_1), \Phi(y_2)) &= \sup_{x \in I[x_0, \delta]} |\Phi(y_1)(x) - \Phi(y_2)(x)| \leq K\delta d_\infty(y_1, y_2). \end{aligned}$$

Hence, the mapping Φ is a contraction since $K\delta \in (0, 1)$ and by the contraction mapping theorem, there exists a unique $y \in X$ such that $\Phi(y) = y$, which completes the proof of the theorem. ■

The contraction mapping theorem is also used in the proof of the implicit function theorem, which another important topic to be covered in these lectures.

In some situations, it is suitable to rather obtain existence results through the following corollary of Theorem 0.4. We recall the notation for the iterates of a mapping $f : (X, d) \rightarrow (X, d)$. Let $n \in \mathbb{N}$. Then

$$f^n := \underbrace{f \circ \dots \circ f}_{n\text{-times}}.$$

Corollary 0.5 *Let (X, d) be a complete metric space and let $f : (X, d) \rightarrow (X, d)$ be a mapping such that f^n is a contraction for $n \geq 2$ integer. Then, the mapping f has a unique fixed.*

Proof: The proof is left as an exercise to be included in the tutorials. ■

The local existence result in Theorem 0.4 above could also be obtained through Corollary 0.5.

0.3 Compactness in metric spaces

Definition 0.6 (Total boundedness)

Let (X, d) be a metric space. We recall that A subset E of X is *totally bounded* if for every $\varepsilon > 0$, there exists $x_1^\varepsilon, x_2^\varepsilon, \dots, x_n^\varepsilon \in X$ such that

$$E \subset \bigcup_{i=1}^n B_X(x_i^\varepsilon, \varepsilon).$$

Exercises 0.7

1. Show that $E \subset X$ is totally bounded then E is also bounded. What about the converse?
2. Show that if $E_1, \dots, E_n \subset X$ are totally bounded then $\bigcup_{i=1}^n E_i$ is also totally bounded.
3. Show that $E \subset X$ is totally bounded if and only if for every $\varepsilon > 0$, there exists a finite number of subsets $E_1, \dots, E_n$ of E with $\text{diam}(E_i) < \varepsilon$ such that $E = \bigcup_{i=1}^n E_i$.

Definition 0.8 (Sequential compactness - Compactness)

Let (X, d) be a metric space.

- (i) We say that X is sequentially compact if for any sequence in X , there exists a subsequence which converges in X .
- (ii) We say that X is compact if for any open covering of X (that is, any family of open sets $(O_i)_{i \in I}$ such that $X = \bigcup_{i \in I} O_i$), there exists a finite subcovering, that is, there a finite subfamily $\{O_{i_1}, \dots, O_{i_n}\}$ such that $X = \bigcup_{j=1}^n O_{i_j}$.

Remark 0.9

- (a) The notions of compactness and sequential compactness coincide in the context of metric spaces. In Mathematical Analysis, (i) is found to be more practical. Therefore, we will often use (i) to show that a given subset of a metric space is compact.
- (b) A subset K of X is compact if $(K, d|_{K \times K})$ is a compact metric space. That is, for any sequence in K , there exists a subsequence which converges to a point in K .
- (c) Let (X, d_X) and (Y, d_Y) be two metric spaces and $f : X \rightarrow Y$ be a continuous mapping. If $K \subset X$ is compact, then $f(K)$ is compact in (Y, d_Y) .
- (d) If $K \subset X$ is compact, then K is closed and bounded. In general, the converse is not true.
- (e) If $K \subset X$ is compact, then K is also complete and totally bounded. Explain why! The converse is also true. In fact, $K \subset X$ is compact if and only if K is complete and totally bounded. (The proof should be found in your 3MS notes).
- (f) We also recall that $E \subset X$ is relatively compact if its closure $\overline{E}$ is a compact subset of X . If $E_1, \dots, E_n \subset X$ are relatively compact then $E = \bigcup_{i=1}^n E_i$ is also relatively compact.
- (f) What about compactness in Euclidean spaces or metric spaces involving finite-dimensional vector spaces? (This should also be found or mentioned in your 3MS)

0.3.1 Sequences of functions between metric spaces

Definition 0.10 (Convergence of sequence of functions) Let X be a non-empty set and let (Y, d_Y) be a metric space.

- (a) We say that a sequence of functions (f_n) from X to Y converges pointwise to the function $f : X \rightarrow Y$ if for every $x \in X$ the sequence $(f_n(x))$ converges to $f(x)$ in the metric space (Y, d_Y) .
- (b) We say that a sequence of functions (f_n) from X to Y converges uniformly to the function $f : X \rightarrow Y$ if for every $\varepsilon > 0$ there exists a number $n_\varepsilon \in \mathbb{N}$ such that

$$n > n_\varepsilon \quad \Rightarrow \quad d_Y(f_n(x), f(x)) < \varepsilon \quad \forall x \in X.$$

That is,

$$(f_n) \text{ converges uniformly to } f \Leftrightarrow \lim_{n \rightarrow \infty} d_\infty(f_n, f) = \lim_{n \rightarrow \infty} \sup_{x \in X} d_Y(f_n(x), f(x)) = 0.$$

The uniform convergence trivially implies the pointwise convergence but the converse is not always true. (Give an example!)

Remark 0.11

- (a) If (f_n) is a sequence of continuous functions from (X, d_X) to (Y, d_Y) which converges uniformly to the function f , then f is also continuous.
- (b) If (X, d_X) is a compact metric space, then the set of continuous functions $\mathcal{C}(X, Y)$ from X to Y is also a metric space when equipped with the distance

$$d_\infty(f, g) := \sup_{x \in X} d_Y(f(x), g(x)) \quad \forall f, g \in \mathcal{C}(X, Y).$$

Moreover, if the metric space (Y, d_Y) is complete, then also $(\mathcal{C}(X, Y), d_\infty)$ is a complete metric space.

Example 0.12 Let (f_n) be a sequence of functions defined by

$$f_n(x) := \frac{x^3}{(1+x^2)^n} \quad \forall x \in [0, \infty), \quad \forall n \in \mathbb{N}.$$

- (a) Show that the sequence (f_n) converges pointwise.

For every $x \in [0, \infty)$, we want to find the limit $\lim_{n \rightarrow \infty} f_n(x)$.

Notice that $f_n(0) = 0$. On the other hand, for every $x > 0$, we have

$$(1+x^2)^n > 1 \quad \Rightarrow \quad \lim_{n \rightarrow \infty} (1+x^2)^n = \infty \quad \Rightarrow \quad \lim_{n \rightarrow \infty} f_n(x) = 0.$$

Therefore, the sequence (f_n) converges pointwise to the function $f \equiv 0$.

- (b) Show that the sequence (f_n) converges uniformly.

Here, we want to show that

$$\lim_{n \rightarrow \infty} \sup_{x \in [0, \infty)} |f_n(x) - 0| = \lim_{n \rightarrow \infty} \sup_{x \in [0, \infty)} \frac{x^3}{(1+x^2)^n}.$$

Let us study the function $f_n(x) = \frac{x^3}{(1+x^2)^n}$ (continuously differentiable).

Notice that, for every $n \geq 2$, we find that $\lim_{n \rightarrow \infty} f_n(x) = 0$.

$$f'_n(x) = \frac{3x^2(1+x^2)^n - 2nx^4(1+x^2)^{n-1}}{(1+x^2)^{2n}} = \frac{3x^2(1+x^2) - 2nx^4}{(1+x^2)^{n+1}} = \frac{x^2[3 + (3-2n)x^2]}{(1+x^2)^{n+1}}.$$

Notice that

$$f'_n(x) = 0 \quad \Leftrightarrow \quad x = 0, \quad \sqrt{\frac{3}{2n-3}}$$

and

$$\left\{ \begin{array}{l} f'_n(x) \geq 0 \quad \Leftrightarrow \quad 0 \leq x \leq \sqrt{\frac{3}{2n-3}} \\ f'_n(x) \leq 0 \quad \Leftrightarrow \quad x \geq \sqrt{\frac{3}{2n-3}} \end{array} \right. \Rightarrow f_n \text{ has maximum at } x = \sqrt{\frac{3}{2n-3}}.$$

Thus,

$$\sup_{x \in [0, \infty)} \frac{x^3}{(1+x^2)^n} = f_n \left(\sqrt{\frac{3}{2n-3}} \right) = \left(\sqrt{\frac{3}{2n-3}} \right)^{3/2} \frac{1}{\left(1 + \frac{3}{2n-3}\right)^n} < \left(\sqrt{\frac{3}{2n-3}} \right)^{3/2}.$$

Thus,

$$\lim_{n \rightarrow \infty} \sup_{x \in [0, \infty)} |f_n(x) - 0| = \lim_{n \rightarrow \infty} \sup_{x \in [0, \infty)} \frac{x^3}{(1+x^2)^n} = 0.$$

Therefore, (f_n) converges uniformly to $f \equiv 0$.

Exercise 0.13 Consider the sequence (f_n) be a sequence of functions defined by

$$f_n(x) := \begin{cases} x & \text{if } 0 \leq x \leq 1/n, \\ 1-x & \text{if } 1/n < x \leq 1, \end{cases} \quad \forall x \in [0, 1].$$

Study the pointwise convergence also the uniform convergence of the sequence (f_n) .

The following proposition by the famous italian mathematician (and politician) Ulisse Dini (1845-1918) gives sufficient conditions under which the pointwise convergence implies the uniform convergence.



Figure 2: **Ulisse Dini:** Italian mathematician, 14 November 1845 - 28 October 1918, with an important contribution in Mathematical Analysis. For instance, in the study of sequences of functions and also in the implicit function theorem (to be covered later).

Theorem 0.14 (Dini's Theorem)

Let (X, d_X) be a compact metric space and $(f_n) \subset \mathcal{C}(X, \mathbb{R})$. Suppose that (f_n) is monotone (increasing or decreasing) and converges pointwise to the continuous function f . Then (f_n) converges uniformly to f .

Proof: Assume for instance that the sequence (f_n) is increasing. For every $x \in X$, $f(x) = \lim_{n \rightarrow \infty} f_n(x)$. So for every $\varepsilon > 0$, there exist an integer $n_\varepsilon(x)$ such that

$$n > n_\varepsilon(x) \quad \Rightarrow \quad |f_n(x) - f(x)| < \varepsilon.$$

Since the sequence (f_n) is increasing and for every $x \in X$, $f(x) = \lim_{n \rightarrow \infty} f_n(x)$, we have that $f_n(x) \leq f(x)$. Set for every $\varepsilon > 0$ and every $n \in \mathbb{N}$,

$$O_n^\varepsilon := \{x \in X : f(x) - f_n(x) < \varepsilon\}.$$

$(O_n^\varepsilon)_n$ is a increasing sequence of open subsets (explain why!) with

$$\left\{ \begin{array}{l} X = \bigcup_{n \in \mathbb{N}} O_n^\varepsilon \\ X \text{ is compact} \end{array} \right. \Rightarrow \exists n_1, \dots, n_k \in \mathbb{N}: \quad X = \bigcup_{j=1}^k O_{n_j}^\varepsilon.$$

Set $m := \max\{n_{j_1}, \dots, n_{j_k}\}$. Since the sequence $(O_n^\varepsilon)_n$ is increasing, it follows that $X = O_m^\varepsilon$ and hence for every $n > m$ we have $X = O_n^\varepsilon$. That is,

$$n > m \quad \Rightarrow \quad \forall x \in X, \quad |f(x) - f_n(x)| < \varepsilon$$

which shows that the sequence (f_n) converges uniformly to f . ■

Notice that we could go deeper in the revision on Metric Spaces with more on compactness, specially applied to the space of continuous functions to prove the [Ascoli-Arzelà theorem](#) used for instance, to establish existence results in the study of qualitative properties of Differential Equations. We could also revisit the notion of connectedness. But we do not want the course to be only on revision!